HelixCode — Fixed Items Summary

> Generated from `docs/Fixed.md` per Constitution §11.4.19. Counts only.
>
> **Round 189 prefix convention:** Open- and closed-item IDs are now scope-prefixed (`HXC`, `HXA`, `HXL`, `HXQ`, `VEN`, …). See `docs/Issues.md` "Prefix convention" section for the full table + legacy `ISSUE-NNN` mapping. The aggregate counts below are unaffected by the rename — only labels change.

Aggregate counts (post round-system rebaseline 2026-05-19)

Type	Count	Closure vocabulary (CONST-057)
Bug	19	`Fixed (→ Fixed.md)`
Feature	68	`Implemented (→ Fixed.md)`
Task	8	`Completed (→ Fixed.md)`

Total closed items: 97 (in the round-system tracker; pre-round closures tracked separately in `docs/improvements/PROGRESS.md`). Round 464 added HXC-010 (End-to-end Kimi CLI + Qwen Code CodeGraph verification — previously operator-blocked on LLM backend quota/credentials, now `Completed (→ Fixed.md)`: operator supplied OpenAI-compatible router credentials; §11.4.10.A pre-use leak-audit CLEAN — `git grep` + `git log -S` confirmed no key value ever committed; the originally-blocking backends remain blocked — `KIMI_API_KEY` shares the same exhausted account-level monthly quota as Kimi's bundled OAuth, Qwen's OAuth free tier still discontinued, OpenRouter credit insufficient — so both agents were driven against the SiliconFlow OpenAI-compatible router: Kimi CLI via an `openai_legacy` provider whose api_key is env-overridden by `OPENAI_API_KEY` (config-file carries only a placeholder), Qwen Code via `--auth-type openai` with env vars never written into the tracked `.qwen/settings.json`; both Challenge scripts updated to honour `HELIX_CG_OPENAI_*` env vars for the credentialed path; result both true tier-1 PASS — each transcript shows the MCP loader connecting to the `codegraph` server with 9 `codegraph_*` tools, the agent invoking `codegraph_search` and the tool returning 10 real `.go` symbol paths from the scanned HelixCode graph; evidence under `docs/research/codegraph/evidence/hxc010/`, secret-scan confirms no key value present; all 7 CodeGraph anti-bluff Challenges now true-end-to-end verified across Claude Code, OpenCode, Crush, Kimi CLI, Qwen Code). Round 463 added HXC-003 (CONST-046 i18n migration backlog — `Implemented (→ Fixed.md)`: the CONST-046 i18n migration campaign is concluded. The genuine user-facing (C) string-literal surface — UI text, prompts shown to the operator, error messages surfaced to end users, labels, helper text — is exhausted across all 7 CONST-046 scope areas: helix_code `internal/` + `cmd/` + `applications/` (exhausted rounds 461/462), LLMsVerifier (round 452), helix_qa, and all owned `vasic-digital/*` + `HelixDevelopment/*` submodules (rounds 413/441). Across ~91-462 rounds tens of thousands of literals were migrated through `nicksnyder/go-i18n/v2` Bundle/Localizer seams plus locale-aware resource files, and every migration round shipped paired-mutation anti-bluff tests that plant a known un-migrated literal and assert the audit-gate reports FAIL — so a PASS certifies real i18n coverage. The audit-gate `scripts/audit-const046-hardcoded-content.sh --fail-on-new` is enforced and each round re-ran `--update-baseline` so the snapshot shrank monotonically. The remaining ~55k audit-baseline hits are all OUT of CONST-046 scope — LLM prompt templates, wrapped-error developer-facing tech strings, identifier tokens, struct-tag keys, format-spec tokens, test fixtures — classified file-by-file in `docs/audits/2026-05-20-internal-const046-classification.md`. CONST-046's invariant is satisfied: no user-facing text is hardcoded as a static literal). Round 403 added HXC-008 (CONST-055 G1 governance gaps surfaced by the post-constitution-pull validation sweep — two pre-existing cascade-drift defects: (a) `docs/improvements/submodule_owned.txt` line 10 referenced a non-existent `dependencies/HelixDevelopment/Models` path which the CONST-052 batch-1 rename `a1ea3c8` had lowercased to `models`, corrected to `dependencies/HelixDevelopment/models` — `verify-governance-cascade.sh` re-run dropped the false `path does not exist on disk` FAIL; (b) `helix_qa/CONSTITUTION.md` was missing anchors CONST-047..057 and `VisionEngine/CONSTITUTION.md` was missing §11.4.69 — both CLAUDE.md/AGENTS.md already carried them, so the CONST-047..057 block was cascaded into `helix_qa/CONSTITUTION.md` from the meta CONSTITUTION.md and §11.4.69 into `VisionEngine/CONSTITUTION.md`; `verify-governance-cascade.sh` → 0 failures, `verify-all-constitution-rules.sh` → 6 gates / 0 failures; helix_qa `1364d23` + VisionEngine `b3a13d8` pushed to all upstreams), HXC-007 (Constitution §11.4.68/70-74 cascade verified genuinely complete — constitution `584b3ee→34a82b3`, 6 rules cascaded to the meta-repo + 67 owned submodules, meta `.gitmodules` pointer confirmed at `34a82b3`), and HXC-009 (owned-submodule GitHub↔GitLab mirror-divergence reconciliation verified complete — helix_qa, VisionEngine, LLMProvider, challenges, containers, DocProcessor all reconciled via merge-first per CONST-061/§11.4.71, all converged + pushed to all upstreams, meta pointers bumped `68309b8e`). Round 402 added HXC-011 (helix_qa runner emitted hollow sub-microsecond `PASSED` rows for desktop-platform bank cases instead of executing the case's `action:` — §11.4 / CONST-035 PASS-bluff in the QA runner itself; reproduce-before-fix Rule 7 / §11.4.43 TDD-fix; root cause: `definitionChallenge.Execute` unconditionally returned `Status=Skipped` and never shelled out, and the generic `challenges/pkg/bank` loader drops each case's `steps` array so executable action data never reached the orchestrator; RED-with-fix-reverted captured the bluff — a desktop `shell:` sentinel-write action ran through `helixqa run -platform desktop` and the sentinel was never written + a `shell: exit 17` case produced 0 failures; fix all in `helix_qa` keeping the challenges submodule decoupled per CONST-051(B) — new `testbank.ActionTypeShell` runs a host command via `os/exec`, `loadExecutableCases()` re-parses banks through `pkg/testbank` which preserves `steps`, `runPlatform` builds a per-platform registry of `definitionChallenge` wrappers carrying the executable case + target platform, and `Execute` on `desktop` runs each `shell:` step capturing the real exit code + output and scores PASS only when every step exits 0 — non-zero scores FAIL, no-executable-action cases emit an honest SKIP with explicit reason per §11.4.3; GREEN: 3 HXC-011 tests PASS, full `pkg/...` suite 122 ok / 0 FAIL, `go build`/`go vet` clean, codegraph bank through the fixed runner 2/2 PASSED real durations + failing case 0/1 FAIL; helix_qa commit `6b46df0`). Round 401 added HXC-012 (data race in `helix_code/internal/llm/load_balancer.go` background stat-collector goroutine resolved via reproduce-before-fix `-race` test — Rule 7 / §11.4.43 TDD-fix; `SelectOptimalProvider` read `lb.currentStrategy` / `lb.strategies` on the hot path without holding `lb.mutex` while `SetStrategy` writes under the mutex and the background `collectStats` goroutine mutates `lb.stats`; fix snapshots the strategy under `lb.mutex.RLock()` then releases before invoking the strategy and `updateStats`, observable behaviour unchanged; new `internal/llm/load_balancer_race_test.go` FAILed RED then PASSes GREEN, full-package `go test -race ./internal/llm/...` PASS, commit `9d8c1cdc`). Round 400 added HXC-006 (HelixCode Speed Programme — all 6 phases / 31 tasks landed making HelixCode 3-5× faster than competitor AI CLI agents; CONST-048 coverage ledger `docs/research/speed/05-coverage-ledger.md` = 29 PASS + 2 honestly-bounded PARTIAL [P5-T01 build/test parallelism tuning landed but suite-wall-time delta not captured as a pasted benchmark; P5-T02 partial single-provider `internal/llm` split — Cerebras extracted, full 18-provider split infeasible without an import cycle] + 0 DEFERRED; headline measured wins P1-T02 lazy Ollama 67µs→2.7ns / P1-T03 lazy CLI startup ~8.85× / P2-T03 repo-map cache ~10.6× warm / P2-T06 incremental tree-sitter ~21× / P3-T03 fast-apply ~516× / P4-T01 PGO −46% CPU-bound, every number carrying pasted in-session captured evidence per CONST-035; release-gate sweep — anti-bluff smoke clean, CONST-046 audit no new hardcoded content, governance-cascade verifier's 2 failures are the pre-existing HXC-008 drift not speed regressions; two pre-existing defects surfaced filed honestly as HXC-011 + HXC-012). Round 396 added HXV-003 (LLMsVerifier `ProviderAdapterForBenchmark.Complete` CONST-050(A) production mock-bluff resolved — WIRE-not-delete decision: the adapter is live code wired by `BenchmarkSystem.Initialize`, so honest deletion was not applicable; the `provider` field was retyped from an unused `interface{}` to the real `benchmark.LLMProvider` interface, `Complete` now dispatches to `a.provider.Complete` returning genuine response text / token count / error, and an honest `ErrProviderAdapterNotWired` sentinel is returned when no real provider is wired — the hardcoded `return "Response", 50, nil` BLUFF-001 fabrication is removed; reproduce-before-fix `httptest` real-dispatch test asserts the adapter actually hits an OpenAI-shim server and returns the real payload; `go test ./internal/benchmark/...` PASS, `go build ./...` clean). Round 397 added OPS-001 (LLMOps 2 pre-existing `CreatePromptExperiment` test failures resolved — both classified (A) test-assertion drift: `ControlPromptCreateFails/TreatmentPromptCreateFails` asserted pre-idempotency behaviour, expecting a same-(name,version) duplicate to fail `CreatePromptExperiment`, but commit `bb53c38` deliberately made prompt registration idempotent so duplicates are tolerated as no-ops; no production regression; tests re-keyed to the genuine non-tolerated failure path — structurally-invalid prompt with empty `Content` → registry `"prompt content is required"` → wrapped with `"control prompt"/"treatment prompt"` prefix; `go test -count=1 ./llmops/...` 2 FAIL → 0 FAIL, `go build ./...` clean). Round 348 added HXV-002 (LLMsVerifier `verification/` package 10 pre-existing test failures resolved — all 10 classified (A) test-assertion drift: 8 `verification_test.go` Verify tests re-keyed to the round-17 honest `ErrVerificationNotWired` contract, 2 `code_verification_test.go` tests re-keyed to honest API-error-propagation / zero-response→`failed` contracts; no production code changed; `go test ./verification/...` 10 FAIL → 0 FAIL, `go build ./...` clean). Round 344 added HXQ-002 (helix_qa `pkg/autonomous` ↔ VisionEngine `remote` API drift resolved — mechanical 3-signature drift: `ProbeHosts/SelectStrongestModel/PlanDistribution` updated to the round-340 VEN-002 merged superset API, now error-returning with `[]SSHConfig/[]ModelSpec` parameters; 5 config-injected pipeline fields added per CONST-045/046; `pkg/autonomous` build+test PASS, helix_agent `tests/integration` compiles). Round 342 added HXA-002 (helix_agent debate/llmprovider sibling-submodule API drift resolved — investigation per operator's explicit ask found the learning/knowledge/recommendations capability tier was GENUINELY DELETED, not moved: `digital.vasic.debate` was rebuilt from scratch at commit `196d0ea` and the slim `CreateDebate/GetStatistics` API is the first+only version, with zero surviving copies anywhere in `dependencies/`; Part-1 import swap + Part-2 slim-API test rewrite + score-scale fix; `debate_integration` tests PASS, anti-bluff per CONST-035). Round 340 added VEN-002 (VisionEngine `vasic-digital-github` divergent fork lineage resolved via CONST-061 §11.4.41 merge-first — real 2-parent merge commit `70c9e0c`, NO force-push, 16-file conflict surface resolved preserving the anti-bluff truth per CONST-035; 7 packages build+test PASS; 4 remotes fast-forwarded). Round 325 added HXQ-001 (helix_qa `TestPerformance` flake resolved — 3 `pkg/vision/` perf tests gated behind `HOST_LOAD_DEDICATED` env var; resolution path (b), timing tolerance NOT loosened, anti-bluff strictness preserved per CONST-035).

Coverage by round-system phase

Phase	Closed items	Status
Round 37-89 (governance + stabilization + LLM wiring)	6 batched closures	Done
CONST-046 Phase 1 (rounds 91-93)	3 (pkg/i18n core + audit script + injection wiring)	Done
CONST-046 Phase 2 (rounds 94-96)	3 (SelfImprove + HelixLLM + harmony_os, 15 strings)	Done
CONST-046 Phase 3 (rounds 97-99)	4 closed (DocProcessor + Planning + VisionEngine + panoptic + audit-gate; 20 strings + audit-gate)	DONE
CONST-046 Phase 4 (rounds 100+)	concluded — campaign closed round 463 (HXC-003 `Implemented (→ Fixed.md)`); genuine user-facing (C) surface exhausted across all 7 scope areas; ~91-462 rounds, tens of thousands of literals	Done

Open issues snapshot (cross-ref `docs/Issues_Summary.md`)

1 open issue: 0 Bug / 1 Task (HXC-001) / 0 Feature; 0 BLOCKED.

Last regenerated: 2026-05-21 (round 464 — HXC-010 closed `Completed (→ Fixed.md)`: operator supplied OpenAI-compatible router credentials; both CodeGraph per-agent Challenges `cg-challenge-05-kimi.sh` + `cg-challenge-07-qwen.sh` re-run produce true tier-1 PASS — Kimi CLI and Qwen Code each genuinely invoke `codegraph_search` and receive real graph data from the scanned HelixCode code-graph; API keys injected via environment variables only, never written to any tracked file; all 7 CodeGraph anti-bluff Challenges now true-end-to-end verified across Claude Code, OpenCode, Crush, Kimi CLI, Qwen Code. Previous round 463 — HXC-003 closed `Implemented (→ Fixed.md)` (CONST-046 i18n migration campaign concluded). See `docs/Fixed.md` for full closure entries with commit SHAs.